Venkata Lakshmi Parimala Pasupuleti

GTM & Analytics Engineer · Python · SQL · Salesforce · Clay · n8n · ETL Pipelines · Deterministic AI · CRM Automation · Real Estate Finance · AWS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA — Open to Denver Hybrid

PROFESSIONAL SUMMARY

Analytics engineer with 8 years building data pipelines that drive business outcomes — not just internal infra. At Entrata: **Salesforce integrations, reporting automations, data sync, and API connections** linking product data to commercial systems; **Clay and n8n workflow automation pipelines**; **200+ SQL query optimizations** (30% performance gain); ETL/ELT pipelines across 20+ production modules serving 100K+ MAU. Expert in **Python, SQL, and relational data modeling** for financial data at scale. Built **Leasel**: deterministic AI pipeline extracting **schema-compliant JSON from unstructured documents** (93.6% precision, 1,240 PDFs) — the same structured prompt engineering + validation pattern needed to sync call recordings to Salesforce. Achieved **100% PHI/PII compliance** with Libsodium encryption — zero security incidents. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 – Aug 2025

Research Associate — Analytics Engineering · ETL Pipelines · Data Modeling

- Built multi-source ETL pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; designed analytics dashboards (Tableau, Quarto) driving **40% improvement** in decision comprehension for faculty leadership.

Entrata — Real Estate PropTech SaaS

Sep 2018 – Aug 2024

Senior Software Engineer — GTM Pipelines · Salesforce · Clay · n8n · SQL · ETL · CRM Automation · AWS

- Built **Salesforce integrations, reporting automations, data sync pipelines, and API connections** linking core product data tables to commercial systems; configured custom objects and automated workflows; built **Clay and n8n enrichment pipelines** connecting product usage data to GTM triggers and replacing manual spreadsheet tracking.
- Designed and owned **ETL/ELT pipelines, schema normalization, and financial data models** across 20+ production modules (100K+ MAU, 99.5% uptime, 120+ Agile sprints); **optimized 200+ SQL queries** (30% performance gain) via execution plan tuning, indexing, partitioning, caching, triggers, log tables, and audit tables; led database architecture decisions for complex production schemas.
- Achieved **100% PHI/PII compliance** (SOC2-aligned) on AWS S3 migration with Libsodium encryption — encrypted 100% of resident PII (SSNs, phone numbers), zero security incidents; eliminated 80% manual workload via Python/Bash automation pipelines; **300+ code reviews, 5+ engineers mentored**.

Entrata India

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · Data Pipelines · API Integrations · AWS S3

- Built full-stack property management modules (PHP, PostgreSQL, JavaScript); designed SQL import scripts for 1.6M HUD/Tax Credit records; developed API integrations, data normalization pipelines, and AWS S3 document storage integrations; participated in sprint planning, code reviews, and QA testing.

PROJECTS

Leasel: Deterministic AI - Schema-Compliant JSON from Unstructured Documents

DETERMINISTIC AI · STRUCTURED OUTPUTS

Python · LangGraph · ChromaDB · Deterministic Rules Engine · Structured Prompt Engineering · JSON Validation — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision** on 1,240 residential lease PDFs — structured prompt engineering extracts **schema-compliant JSON from unstructured legal documents**; deterministic rules engine validates every LLM output before writing to database, ensuring fully auditable, corruption-free records. Same pattern applies directly to call recording → Salesforce field sync pipelines.

AgentProbe: Multi-Provider AI Pipeline with Observability & CI/CD Gates

AI ENGINEERING · DATA PIPELINES

Python · LangGraph · FastAPI · ChromaDB · Docker · GitHub Actions · React — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime with multi-provider LLM routing (Claude, OpenAI, Groq, Ollama), CI/CD regression gates on every commit; React **observability dashboard** tracking per-run metrics and retrieval drift; demonstrates end-to-end pipeline engineering from data ingestion through validation to production monitoring.

CrashClarity: Analytics Engineering on 104K Records with Attribution Dashboard

ANALYTICS ENGINEERING

Python · R · scikit-learn · Random Forest · Feature Engineering · ETL · Tableau · GitHub Pages

- **85%+ accuracy** on 104K+ records — ETL pipeline, schema normalization, feature engineering, multi-model comparison, and **interactive attribution dashboard** on GitHub Pages tracking outcomes across multiple signal dimensions.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

Data engineering, ML systems, statistical modeling, AI pipelines, data governance, anomaly detection

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

GTM & CRM Engineering: Salesforce (integrations, reporting automations, data sync, API connections, custom objects), Clay (enrichment pipelines), n8n (workflow automation), Zapier, CRM data pipelines, lead routing, billing automation, multi-touchpoint attribution reporting, PII/GDPR/CCPA compliance, SOC2-aligned data handling

Data Engineering & Backend: Python (expert), SQL (expert — 200+ production optimizations, schema design, normalization, execution plan tuning, indexing, partitioning, triggers, audit tables), PHP, JavaScript, TypeScript, R; ETL/ELT Pipelines, Schema Design, Data Normalization, Data Validation, API Integrations (REST), AWS (S3, EC2, RDS, Glue), PostgreSQL, MySQL, Docker, CI/CD (GitHub Actions), Git, Linux

AI & Deterministic Outputs: Structured Prompt Engineering, Deterministic Rules Engine, Schema-Compliant JSON from Unstructured Data, LangGraph, LangChain, RAG Pipelines, ChromaDB, LLM Evaluation, FastAPI, Claude API, OpenAI API, Groq; ML: PyTorch, scikit-learn, Isolation Forest, Feature Engineering, Tableau, Power BI